

Keywords

Particle physics, astrophysics, cosmology;

(Light and Ultralight) Dark Sectors, Dark Photons, Inelastic Dark Matter, Direct Detection, Self Interacting Dark Matter, Big Bang Nucleosynthesis & New Physics

Education & Academic Career

2024–now **Postdoc**, *Sapienza University of Roma*, Italy
Part of Andrea Caputo's ERC grant DarkLS

Description Investigation of astrophysical environments and cosmology in the context of new physics: neutron star cooling, plasma effects for resonant dark matter conversion, BBN constraints, Ultralight Dark Matter

2024–2024 **Postdoc**, *DESY Hamburg*, Germany
1 Month extension of the PhD contract for October

2021–2024 **PhD**, *DESY Hamburg & Universität Hamburg*, Germany, GPA – 1.0 (magna cum laude)

Thesis *Catching Dark Photons in the Sky: Looking for light vector particles using satellites*

Supervisor Kai Schmidt-Hoberg

Description Investigation of the Dark Photon model: solar production, Dark Photon Dark Matter searches

2019–2021 **M.Sc. Physics**, *RWTH Aachen*, Germany, GPA – 1.0
Focus of Studies: Quantum Field Theory and Gauge Theories

Thesis *Exothermic Dark Matter from Upscattering in the Earth*

Supervisor Felix Kahlhoefer

Description Construction of a general framework for inelastic dark matter in direct detection searches

2016–2019 **B.Sc. Physics**, *RWTH Aachen*, Germany, GPA – 1.3

Thesis *Self-interacting Dark Matter from the Yukawa Potential: From astrophysical data to particle properties*

Supervisor Felix Kahlhoefer

Description Investigation of Dark Matter self-interactions on the shape of Dark Matter halos

2008–2016 **High school/Abitur**, GPA – 1.0

GPA for Germany: 1.0–4.0 (1.0 is the highest possible score; 5.0 corresponds to not passed)

Publications

- [1] Timon Emken, Jonas Frerick, Saniya Heeba, and Felix Kahlhoefer. Electron recoils from terrestrial upscattering of inelastic dark matter. *Phys. Rev. D*, 105(5):055023, 2022. [arXiv:2112.06930](https://arxiv.org/abs/2112.06930), doi:10.1103/PhysRevD.105.055023.
- [2] Jonas Frerick, Felix Kahlhoefer, and Kai Schmidt-Hoberg. A' view of the sunrise: boosting helioscopes with angular information. *JCAP*, 03:001, 2023. [arXiv:2211.00022](https://arxiv.org/abs/2211.00022), doi:10.1088/1475-7516/2023/03/001.
- [3] Jonas Frerick, Joerg Jaeckel, Felix Kahlhoefer, and Kai Schmidt-Hoberg. Riding the dark matter wave: Novel limits on general dark photons from LISA Pathfinder. *Phys. Lett. B*, 848:138328, 2024. [arXiv:2310.06017](https://arxiv.org/abs/2310.06017), doi:10.1016/j.physletb.2023.138328.
- [4] Sara Bianco, Paul Frederik Depta, Jonas Frerick, Thomas Hambye, Marco Hufnagel, and Kai

Schmidt-Hoberg. Photo- and hadrodisintegration constraints on massive relics decaying into neutrinos. *JCAP*, 11:072, 2025. [arXiv:2505.01492](#), [doi:10.1088/1475-7516/2025/11/072](#).

- [5] Sara Bianco, Jonas Frerick, Marco Hufnagel, and Kai Schmidt-Hoberg. Improved Big Bang Nucleosynthesis constraints on decaying massive relics. 5 2026. [arXiv:2605.26824](#).
- [6] Jonas Frerick, Hyungjin Kim, and Felix Kling. Precision Solar System Dynamics for Ultralight Dark Matter Search. 6 2026. [arXiv:2607.00072](#).

Selected Presentations

Conference talks & posters

- 2022 *iDM@IDM: Electron recoils from terrestrial upscattering of inelastic dark matter*, IDM 2022, Vienna
- 2022 *Light-shining-through The Dark Side of the Moon*, DESY Theory Workshop 2022, Hamburg
- 2023 *A' view of the sunrise: Boosting Helioscopes with Angular Information*, IAXO Meeting, Hamburg
- 2024 *Riding the dark matter wave: Novel limits on general dark photons from LISA Pathfinder (+Poster)*, Invisibles24 Workshop, Bologna
- 2025 *Catching Dark Photons in the Sky - Looking for light vector particles using satellites*, LULBI Workshop, Weizmann Insitute
- 2025 *Photo- and hadron-dissociation constraints on relics decaying into neutrinos*, Dark Matter and Neutrinos, Paris
- 2026 *New BBN constraints on the decay of relics into SM particles*, PLANCK2026 and the 6th EuCAPT Symposium, CERN

Lectures

- 2023 *Quantum Computing V*, DESY Workshop Seminar

Seminars

- 2023 *Looking for Dark Photon Dark Matter with LISA Pathfinder*, Fermilab Theory Chalk Talk
- 2025 *Photo- and Hadrodisintegration constraints on massive relics decaying into neutrinos*, BSM Seminar, KIT (Karlsruhe)
- 2025 *Photo- and Hadrodisintegration constraints on massive relics decaying into neutrinos*, CERN TH Cosmo Coffee

Schools & Visits

- 2023 **Summer School on Particle Physics**, ICTP Trieste
- 2023 **Asymmetry Secondment**, Fermilab
- 2024 **Invitation to the LULBI workshop**, Weizmann Institute
- 2025 **Invited visit by Maksym Ovchinnikov**, CERN

Grants, Fellowships & Awards

- 2024 **Quantum Universe best paper award** (Awarded by the excellence cluster in Hamburg)
- 2023 **Marie Curie Staff exchange program "Asymmetry"** (Funding for 2 months of stay at Fermilab)
- 2021 **Deutschlandstipendium**
- 2020-2022 **Dean's List of the RWTH Aachen University**
- 2016 **DPG Abiturpreis** (Awarded by the German Physical Society DPG)

Coding & Software

- Advanced Python, Mathematica
- Basic Fortran, C, C++
- Software AlterBBN (with personal modifications), ACROPOLIS (with personal modifications), NSCool, PYTHIA, MadGraph